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|---|---|---|
| Examination | : | **End Semester Examination – MAY/JUNE 2026** |
| Name of the Course | : | **B.Tech. (IT & Mathematical Innovations)** |
| Name of the Paper | : | **Numerical Methods for Computational Mathematics** |
| Unique Paper Code | : | **3122613601** |
| Semester | : | **VI** |
| Duration | : | **2 hours** |
| Maximum Marks | : | **60** |

***Instruction to students:***

*This question paper contains two sections, attempt given number of question from both the sections. Use of non-programmable scientific calculator is allowed in the exam.*

## SECTION-A

Attempt any **FIVE** questions from this section, each carries six marks. ($5 \times 6 = 30$)

1. Consider the function $f(x) = 1 + lnx - x$. What is the multiplicity of the root at $x = 1$? What will be the order of convergence of Newton's method to approximate this root? How can we restore the quadratic convergence of Newton's method in this case starting with a initial approximation of $p_0 = 2$? (1+1+4)

2. Determine a formula that relates the number of iterations 'n' required by the bisection method to converge to an absolute error tolerance of $\epsilon$ starting from an initial interval $(a_0, b_0)$. While computing the root of $f(x) = x(1 - cosx)$ on the interval $(-2,1)$, to achieve the accuracy within 6 significant digits, how many iteration do we need to perform? (3+3)

3. The velocity of an object as a function of time is given by:

| Time($t$) | 2 | 5 | 7 | 9 | 12 |
|---|---|---|---|---|---|
| velocity($v(t)$) | 12 | 16 | 24 | 15 | 33 |

Using Simpson's 1/3rd rule, calculate the distance travelled by the object between $t = 2$ and $t = 12$.

4. Show that the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ -1 & 0 & 2 \end{bmatrix}$ has no LU decomposition. Rearrange the rows of A so that the resulting matrix does have an LU decomposition. (3+3)

5. Localise the eigen values of given matrix A using Gerschgorin Circle theorem. Perform **three** iterations of Power method to find **smallest** eigen value and corresponding eigen vector of matrix A. (2+4)

$$\begin{bmatrix} -2 & -2 & 3 \\ -10 & -1 & 6 \\ 10 & -2 & -9 \end{bmatrix}.$$

6. Construct the divided difference table for the following data set and then write out the Newton's form of Interpolating polynomial. Find the value of 'y' at $x = -2$

| | | | | |
|---|---|---|---|---|
| **x** | **-7** | **-5** | **-4** | **-1** |
| **y** | **10** | **5** | **2** | **10** |

*SECTION-B*

Attempt any **THREE** questions from this section, each carries ten marks. $(10 \times 3 = 30)$

1. Let $A$ be a strictly diagonally dominant matrix and let $T_{jac}$ be the Jacobi method iteration matrix associated with $A$. Show that $\rho(T_{jac}) < 1$. Perform two iterations of SOR method with $\omega = 0.8$ to solve the following system of linear equations $AX = b$ with an initial approximation $x^{(0)} = [0 \quad 0 \quad 0]$.

$$\text{where } A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 4 & 2 \\ 0 & 2 & 6 \end{bmatrix} \text{ and } b = \begin{bmatrix} -1 \\ 3 \\ 5 \end{bmatrix}.$$

2. The liquid level in an accumulator for a pumped fluid system satisfies the model;

$$\frac{dh}{dt} + 0.002\left(52.1 \times h + \frac{10.3}{(10.3 + h)}\right) - 1.17(1 + \sin 3t) = 0.0308,$$

$$h(0) = 5.0$$

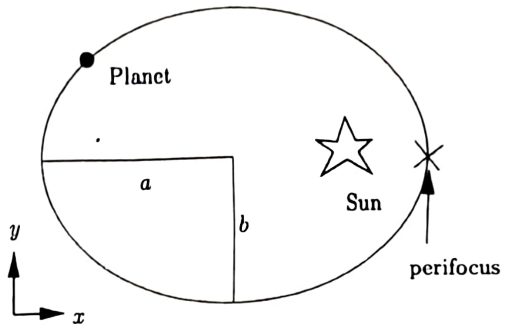
where 'h' denotes the liquid level (measured in meters) and 't' denotes time (measured in minutes). How does the liquid level evolve in time. Use Runge-Kutta method of fourth order and estimate the liquid level after 5 minutes in two iterations.

3. Consider the Kepler equation of motion M = E − e sin E, relating the mean anomaly M to the eccentric anomaly E of an elliptic orbit with eccentricity e. To find E, we need to solve the nonlinear equation: f (E) = M + e sin E − E = 0.

Given e = 0.0167 (Earth's eccentricity)and M = 1 (in radians) compute E using fixed point iteration with an error tolerance of $10^{-4}$ and approximate the position of the planet Earth for t = 5 days with $a = 150 \times 10^6 km$. Given that, the position (x, y) of the planet at time t can be determined by the following formula:

$$x = acos(E - e)$$

$$y = a\sqrt{1 - e^2}sine$$




4. (a) Consider the definite integral $\int_0^2 x^2 e^{-x} dx$. Numerically determine the rate of convergence of the composite two-point Gaussian quadrature rule, taking 2 equal subintervals. (5)

(b) A thermodynamic student needs the temperature of saturated steam under a pressure of a 6.3 mega Pascals (MPa). Estimate the temperature using Lagrange Interpolation from the data; (5)

| **Pressure (MPa)** | 4.0 | 5.0 | 6.0 | 7.0 | 8.0 | 9.0 |
|---|---|---|---|---|---|---|
| **Temperature ($^0C$)** | 250.40 | 263.99 | 275.64 | 285.88 | 295.06 | 303.40 |